

Mohamed Abdelrahman

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EDUCATION

University of Houston

Bachelor of Science in Computer Science - Major GPA: 3.81

Expected Graduation: Dec 2026

Houston, TX

EXPERIENCE

Site Reliability Engineer, Intern

May 2026 – Aug 2026

IBM

Austin, TX (Hybrid)

- Built a rule-based label migration tool in Python integrating Jira's REST API, with an LLM fallback for ambiguous cases, to resolve mislabeled fields across 6,000+ SRE Jira issues, making thousands of hours in team credit trackable
- Built a distributed REST API client with token-bucket rate limiting to coordinate 128 concurrent workers against the LLM API after the team's single-request interface became a bulk-processing bottleneck, sustaining high-throughput access without overloading the backend
- Built a schema-driven, multi-prompt LLM pipeline orchestrating 5 parallel extraction tasks (scope, timeline, root cause, customer impact, RCA) per issue across 300 SRE incident reports, with an independent LLM review pass validating each output, cutting analysis time from 2.8 hours to 10 minutes (17x speedup)
- Built a configurable, diff-based label migration tool with dry-run, rollback, and fail-safe checks for safe, reliable production rollout of label changes

Course Facilitator

Jan 2026 – Present

Department of Computer Science - University of Houston

Houston, TX

- Prepare coding problems and instructional material to facilitate COSC 3320: Algorithms and Data Structures
- Host two workshops per week, coordinating with faculty and facilitators to ensure effective learning experiences

Software Engineer (Consultant)

Nov 2022 – Apr 2023

bld.ai

Remote

- Developed a full-stack platform using React and Django to match families for playdate arrangements
- Refactored database models, authorization, and data validation, reducing bugs in the codebase by 50%
- Deployed a minimum viable product (MVP) within one month using a two-week sprint cycle
- Designed and configured role-based permission groups to control access while ensuring data integrity

LEADERSHIP

Director of Coding Competitions

Jan 2025 – Present

IEEE-NSM - University of Houston

- Prepared and hosted workshops on data structures and algorithms in C++, and curated competitive programming competitions to strengthen undergraduate students' algorithmic problem-solving skills
- Organized practice sessions for the ICPC team by selecting problems and arranging virtual contest sessions, contributing to the team's 1st place finish at UH Site
- Coordinated with students and faculty on ICPC hosting logistics at UH, organizing scheduling shifts for 20 volunteers to ensure successful contest execution

AWARDS

Golden Medalist, ICPC South Regional Contest - UH Site

Nov 2025

University of Houston

- Secured 1st place at UH Site (16th place regionally) in the Division 1 ICPC South Regional Contest by solving 4 problems using C++ and advanced algorithmic techniques

PROJECTS

C++ Ordered Map Library | C++17, CMake, Catch2, GitHub Actions, Test-Driven Development

- Developed high-performance ordered map data structure combining doubly linked list with hash table for O(1) lookups while preserving insertion order, with STL-compatible interface and exception safety
- Implemented comprehensive test suite achieving 97.3% code coverage across 2,246 test assertions with Catch2, established CI/CD pipeline via GitHub Actions, and verified zero memory leaks through Valgrind analysis

TECHNICAL SKILLS

Languages: C++, Python, SQL, JavaScript, HTML/CSS, Dart, C#

Frameworks: Django, React, NextJS, Flutter, Unity

Tools: Git, Docker, Postman, Coverage, GitHub Actions, VS Code, Ubuntu Linux, Bash